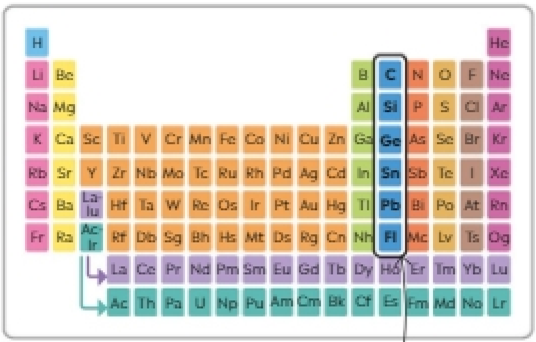


The six elements of the carbon group sit between the boron group and the nitrogen group.

The Carbon Group

Like its neighbors on the periodic table, the carbon group is another mixed bag of elements. While carbon is a non-metal, silicon and germanium are metalloids. Tin and lead are metals that have been used in industry for millennia, while flerovium is an artificial element.

Uncut diamond, a form of carbon

Colorless, crystalline mineral

Carbon

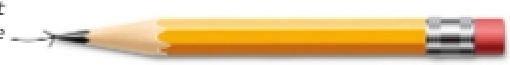
6 C

Element number six is arguably one of the most important of all of the 118 known elements. All organic, living things on Earth contain carbon atoms. Carbon atoms can form chemical bonds with each other to form large molecules and carbon produces millions of compounds, including those important for life—amino acids, fats, and DNA. From fossil fuels to diamonds, plastics, and all living organisms, carbon is everywhere.

Glass-like carbon

This human-made material is called glassy carbon. It has a very high resistance to heat and does not easily react with oxygen. This means it can be used at high temperatures—for example, to make metalworking crucibles that hold molten metals.

Soft graphite



Common carbon

Graphite and diamond are two especially common forms of carbon. Their distinctive atomic structures mean that they have very different properties: graphite is the soft and slippery material used in pencils, and diamonds are ultra-hard precious gemstones also used as a cutting tool in industry.

Fossil bone analysis

Carbon dating

A radioactive form (isotope) of carbon called carbon-14 is used to date ancient materials, such as bones. Over time, carbon-14 in an object breaks down, or "decays." The amount of carbon-14 in a sample is measured and compared with how long it takes for half of the radioactive carbon atoms to decay (the carbon's "half-life"). This gives scientists an estimate of the item's age.

Formula One race car